16. TESTING:

A. PRINTED WIRING BOARD AND TEST COUPON TO BE DESIGNED TO THE REQUIREMENTS OF IPC-2221 CLASS 3 AND TESTED PER IPC-6012 CLASS 3 TYPE 3.

B. ELECTRICAL TEST SHALL BE PERFORMED ON 100% OF THE DELIVERABLE PRODUCT. THE ELECTRICAL PROGRAM SHALL BE EXTRACTED FROM GERBER DATA OR IPC-356 FILE.

C. FOREIGN INCLUSIONS ALLOWED: 2 PER SIDE OF BOARD.

D. LAYER TO LAYER MIS-REGISTRATION ALLOWED: MAX .005. TOTAL OF .014 OVER ALL LAYERS.

E. NO OUTGASSING MASS LOSS TEST REQUIRED.

F. DIELECTRIC WITHSTANDING VOLTAGE (HIPOT) LAYER TO LAYER: 750VDC +15 -0 FOR 30 SECONDS. TO BE DONE ON TEST COUPON.

G. RATING: MAXIMUM RATED VOLTAGE: 28VDC.
TESTING TO BE AT DOUBLE THE MAXIMUM RATED VOLTAGE.

17. TEST COUPONS TO BE INTEGRATED INTO PANEL PER IPC-2221 CLASS 3 ARE TO BE ADDED BY MFG. TEST COUPONS TO BE TRACEABLE TO LOT AND PWB SERIAL NUMBER. TEST COUPON TO BE LOCATED WITHIN .5 INCHES OF ANY BOARD ON THE PANEL.

18. BOARD MANUFACTURER IS TO BE ON THE QPL FOR MIL-PRF-31032/1 OR MIL-PRF-55110.

9. HOLES:
DIAMETERS AND LOCATION TOLERANCES SHALL BE IAW IPC-2222, IPC-2221 CLASS 3 LEVEL B AND ASME Y14.5M-1994.
ALL HOLES ARE LOCATED

⌖	Ø.006 Ⓜ	A	B Ⓜ	C Ⓜ

AND ARE SPECIFIED IN THE ACCOMPANYING DRILL FILE.
FEATURES ARE DEFINED FROM DATUM B.

10. DRILL PER SUPPLIED FILE: S1101EFL0302_REV_B.PTH
S1101EFL0302_REV_B.NPT

11. ELECTRONIC DATA FILES:
GERBER FILES (IN EIA RS-274X) AND IMAGE FILES (PDF) ARE LISTED IN THE README.TXT FILE AND ARE SUPPLIED IN THE APPROPRIATE MEDIA (S1101EFL0302_REV_B_9-15-2014.ZIP)

12. BARE BOARD IDENTIFIED WITH PN AND ARTWORK REVISION. ARTWORK IMAGES ARE VIEWED THROUGH LAYER 1 (PRIMARY SIDE). CURRENT ARTWORK REVISION (CONFIG IDENT LETTER) IS LOCATED ON THE ARTWORK AND SECONDARY (BOTTOM) SIDE OF BOARD IN ETCH.

13. SOLDER MASK:
APPLY LPI (LIQUID PHOTO IMAGEABLE) PERMANENT GREEN SOLDER MASK ON PRIMARY (TOP) AND SECONDARY (BOTTOM) SIDE BOARD OVER BARE COPPER PER IPC-SM-840 CLASS H. FOR SOLDER MASK USE ARTWORK MASTER SUPPLIED, LISTED IN THE README.TXT FILE.

14. FINAL FINISH:
EXPOSED COPPER AREAS SHALL BE PRE-TINNED WITH A SINGLE PASS HOT AIR SOLDER LEVELING (HASL) PROCESS. THE PREFERRED HASL PROCESSING METHOD IS A HORIZONTAL PLANE HOT AIR SOLDER LEVEL UTILIZING A PROCESSING ANGLE OF 45 DEGREES.
PAD TO PAD THICKNESS UNIFORMITY SHALL BE MAINTAINED.
HASL MINIMUM THICKNESS SHALL BE COVERAGE AND SOLDERABLE.
HASL MAXIMUM THICKNESS SHALL BE .001 INCHES.
HASL CO-PLANARITY SHALL BE +/- 1000 MICRO INCHES.

15. MARKING:
SILKSCREEN, LEGENDS AND OTHER NOMENCLATURE ON TOP OF THE SOLDER MASK USING AA56032-IIWHT WHITE EPOXY INK PER MFG INSTRUCTIONS AND MIL-STD-130. FOR SILKSCREEN USE ARTWORK MASTER SUPPLIED, LISTED IN README.TXT FILE.
INDICATE THE FOLLOWING:
MANUFACTURER ID
DATE CODE
SERIAL NUMBER

NOTES: UNLESS OTHERWISE SPECIFIED

1. NO CHANGES BEYOND CUSTOMARY AND STANDARD BUILD PRACTICES SHALL BE MADE WITHOUT PRIOR APPROVAL FROM ZIN TECHNOLOGIES.

2. PRIMARY SIDE OF BOARD SHOWN.

3. FABRICATION SHALL BE IAW IPC-2221 AND IPC-2222.

4. INSPECTION AND ACCEPTANCE CRITERIA PER IPC-6012 CLASS 3 TYPE 3.

5. MATERIAL:

A. GLASS CLOTH EPOXY LAMINATE FLAME RESISTANT PER IPC-4101/24. COPPER ELECTRO DEPOSITED HIGH TEMPERATURE TYPE H, THICKNESS TOLERANCE CLASS B AND SURFACE QUALITY CLASS A. SEE LAMINATE STACK UP FOR NORMAL FOIL WEIGHT, UNLESS OTHERWISE INDICATED, MINIMUM DIELECTRIC SPACING BETWEEN CONDUCTIVE LAYERS SHALL BE .0035 MINIMUM AFTER LAMINATION.

B. BONDING SHEET: RESIN PRE-IMPREGNATED GLASS FABRIC PER IPC-4101/24.

C. COPPER FOIL: ELECTROPLATED COPPER SHEET PER IPC-MF-150, HTE-TYPE E. .5 OZ BEFORE PLATING MAY BE SUBSTITUTED FOR IPC-4101/24 MATERIAL FOR EXTERNAL LAYERS.

6. CONDUCTORS:
DESIGNED MINIMUM ACCEPTABLE CONDUCTOR WIDTHS AND SPACING AS FOLLOWS:

	INTERNAL	EXTERNAL
CONDUCTOR WIDTH ≤ TO .013	+.001 -.002	+.003 -.003
CONDUCTOR WIDTH > TO .013 AND < .025	+.002 -.004	+.004 -.004
CONDUCTOR WIDTH ≥ TO .025	+.004 -.006	+.008 -.006

WITH MINIMUM SPACING OF .008.
MINIMUM CONDUCTOR WIDTH OF .008.

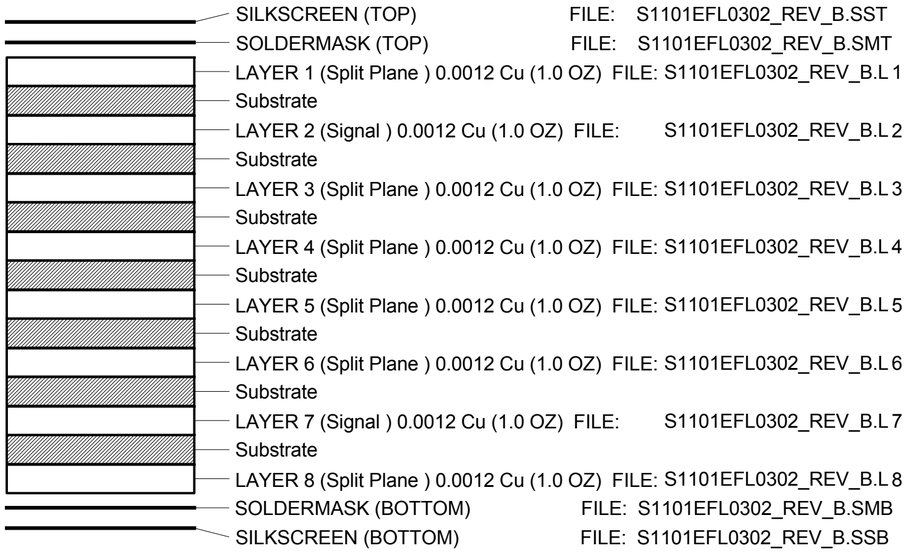
7. MINIMUM INSULATION DISTANCE FROM ANY ACTIVE CIRCUITRY TO THE BOARD EDGE SHALL BE .025.

8. MINIMUM ANNULAR SOLDER RING SHALL BE PER IPC-6012, CLASS 3. THERE SHALL BE A MIN. OF .002 EXTERNAL AND .001 INTERNAL OF COPPER PAD ANNULAR RING, NOT INCLUDING HOLE PLATING MATERIAL, AROUND DRILL HOLES FOR ELECTRICAL CIRCUITS.

A. VIA'S MAY BE PLATED CLOSED.

B. VIA PLATING MAY NOT LEAVE SMALL HOLES OR POCKETS THAT EXPLOSIVELY OUTGAS DURING SOLDER PLATING.

LAYER STACK-UP



REV	DESCRIPTION	APPROVED
C	REVISE PER ACME-ECO-373	MH 3/18/15 TG 3/19/15 DS 3/19/15
B	REVISE PER ACME-ECO-315 AND ACME-ECO-328	TAG 9/16/14 DAS 9/16/14 RJP 9/16/14
A	REVISE PER ACME-ECO-295	TAG 4/14/14 DAS 4/10/14 RJP 4/10/14

REVISION HISTORY

.
S1101EFA0302
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
.X ± 0.03
.XX ± 0.01
.XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 3/19/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 7/30/13	QUALITY	D. CHOVAN 7/31/13
REVIEWER	D. STRANG 7/31/13	MATERIALS	M. WILKINSON 8/5/13
MECHANICAL	A. DRAKE 7/31/13	PROJ LEAD	B. BOROWSKI 8/19/13
ELECTRICAL	T. GOBELI 7/31/13	-	-
STRUCTURAL	W. BORRELLI 7/301/13	-	-
SAFETY	C. WERNER 7/31/13	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE110

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
WWW.ZIN-TECH.COM

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

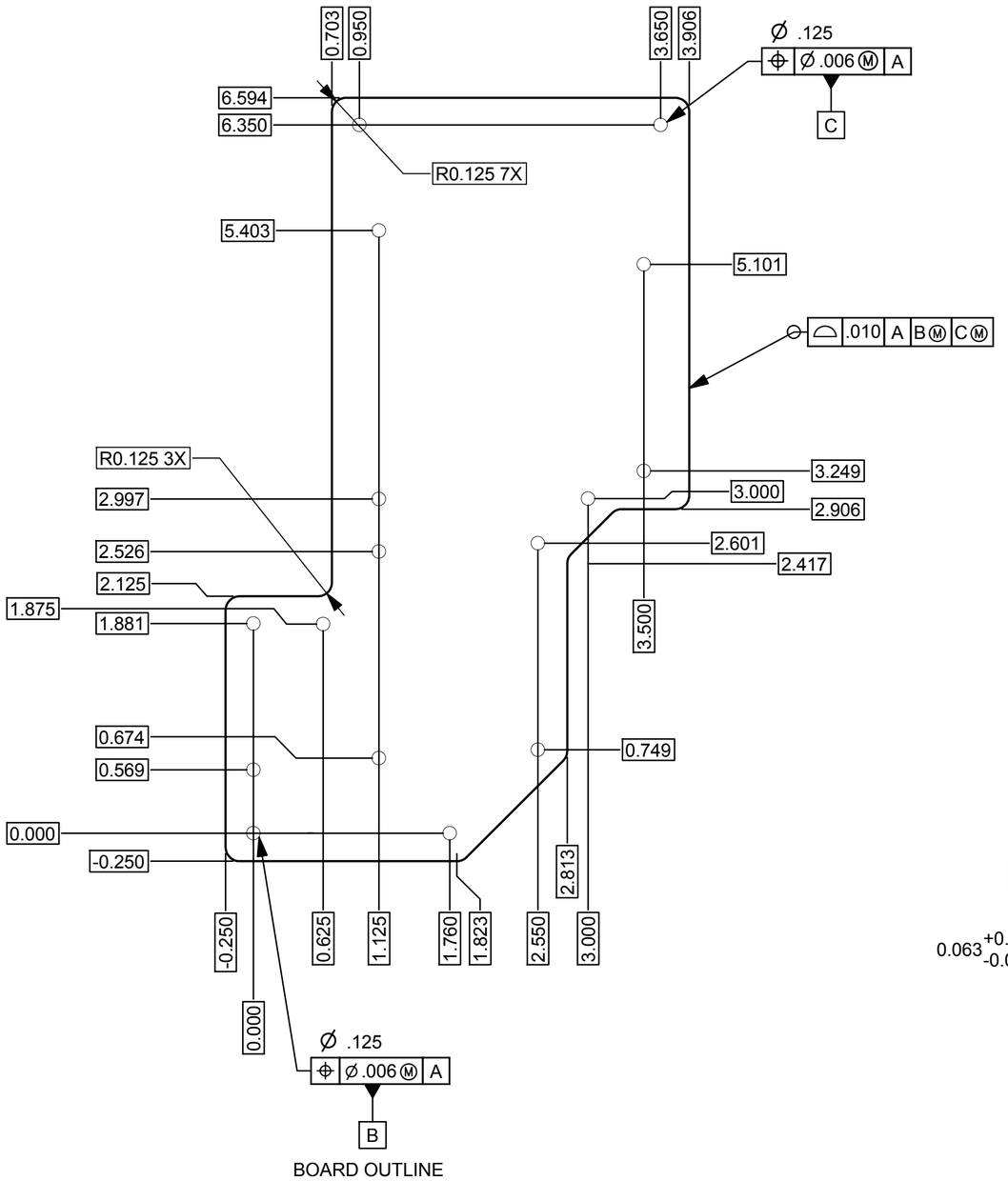
TITLE
ACME
UPSTREAM RELAY CONNECTOR BOARD
FABRICATION DRAWING

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFF0302	C

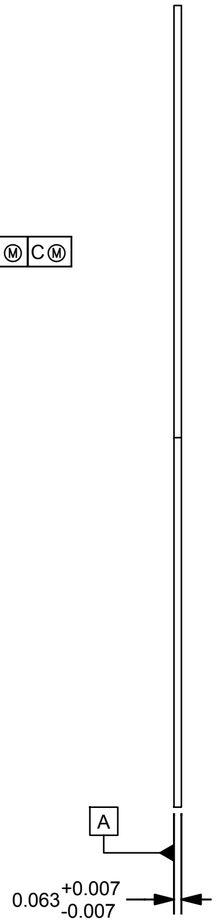
SCALE: NONE | INITIAL REL DATE 8/19/13 | SHEET 1 OF 2

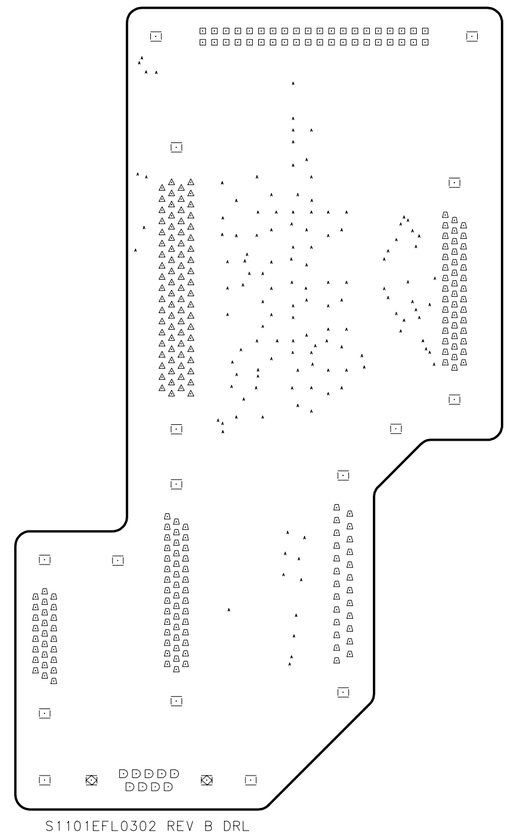
RELEASE LEVEL: .
MODEL NAME: S1101EFA0302_REV_C.bpd
BLUEPRINT
VERSION 4.0

REV C
SH 1
DWG NO S1101EFF0302



BOARD OUTLINE





S1101EFL0302 REV B DRL

Through Holes				
Symbol	Quantity	Diameter	Tolerance	Plated
A	135	0.012	+0.003 / -0.012	Yes
◬	139	0.033	+0.004 / -0.003	Yes
△	78	0.041	+0.004 / -0.003	Yes
⊡	40	0.046	+0.004 / -0.003	Yes
D	9	0.049	+0.004 / -0.003	Yes
⊠	2	0.119	+0.004 / -0.003	Yes
⊡	12	0.125	+0.004 / -0.003	Yes
⊡	4	0.125	+0.004 / -0.001	No

DRILL PATTERN

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFF0302	C
SCALE: NONE	INITIAL REL DATE 8/19/13	SHEET 2 OF 2	

BLUEPRINT VERSION 4.0

REV C
SH 2
DWG NO S1101EFF0302